

Intermediate Project Report

infrared.city-1: Tree detection

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April 18, 2026

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1 Introduction

This section sets the context for the research by outlining the underlying motivation, the proposed methodological approach, and the boundaries of the study. It guides the reader from the broader societal challenge of urban climate adaptation to the specific technical implementation of tree detection using satellite imagery.

1.1 Motivation and Project Overview

As urban areas expand and global temperatures rise due to climate change, being able to understand how trees can help cool cities has become increasingly important [1]. Furthermore, trees reduce the risk of flooding in cities and promote health and wellbeing [2]. Our project aims at taking the first step in the direction of better tree coverage in urban areas by creating a model that is able to predict precise tree locations using satellite imagery of densely populated areas.

This research builds upon the foundation laid by previous projects conducted in collaboration with the partner company and environmental simulation platform *infrared.city*. A recent flood risk analysis developed for the platform highlighted that surface impermeability is a dominant factor in urban flooding [3]. However, the authors noted that the lack of detailed, highly localized environmental data limits the precision of urban risk analyses [3]. Concurrently, a second project successfully mapped general urban green spaces using Random Forest models on Sentinel-2 data [4]. While effective for broad classification, that project grouped all vegetation into a single category and reported that initial attempts to use U-Net architectures for detailed spatial analysis failed to yield good results [4]. The authors explicitly recommended future work to revisit U-Net architectures and to distinguish vegetation types more effectively [4].

Recently, there has been a rapidly growing body of research on urban tree detection using deep learning models. One of the most recent papers is by Gong et al. [5], who use satellite data not only to detect trees but also to estimate their size and density. Our project tries to build on both this literature and the identified shortcomings of the previous *infrared.city* projects. By integrating multi-season imagery and temporal attention algorithms, we aim to overcome the previous limitations of the U-Net model and achieve a more granular, object-specific classification than general green spaces.

1.2 Proposed Approach and Data

To achieve reliable urban tree detection, our approach combines multi-spectral satellite imagery with a deep learning architecture optimized for image segmentation.

We utilize publicly available Sentinel-2 satellite images, providing a 10-meter spatial resolution across four color channels: red, green, blue, and near-infrared. These images are retrieved via the API of the Copernicus Programme and are provided by the European Space Agency [6]. A key technical focus of our data strategy is the inclusion of imagery from different seasons to account for the significant impact of seasonality on tree appearance. To provide ground truth for training and testing, we integrate the publicly available “Baumkataster” dataset, which contains exact locations for every registered tree in the city of Vienna [7].

Regarding the model, we employ a U-Net architecture, a specialized Convolutional Neural Network (CNN) consisting of symmetric contracting and expanding paths [8]. While a U-Net implementation in a previous project did not yield promising results, this paper aims to build upon that foundation as the architecture is considered highly accurate for detecting objects in satellite data [9, 10]. Our model extracts features from seasonal multi-channel imagery through its contracting path and utilizes an expanding path for upsampling to predict precise locations.

To improve upon previous attempts, our implementation incorporates skip connections and a temporal attention algorithm to prioritize more meaningful images within the time series [10]. The model performance is optimized using Binary Cross-Entropy loss, comparing our predictions against the split training and test sets derived from the Vienna “Baumkataster”.

1.3 Scope and Limitations

The model in this paper is trained only on satellite images of urban areas and is therefore only supposed to be used for tree detection within city environments. Additionally, the model focuses on tree detection in the temperate climate zone, as this is the vegetation it is trained to observe. Further analyses on the impact of trees on, for example, heat regulation in cities and flooding prevention are out of the scope of this project and could be the topic of future research.

Since this paper uses Sentinel-2 satellite images with 10-meter spatial resolution, issues could arise, as the images might not have a high enough resolution to detect trees reliably. Future work could therefore also focus on improving image quality by using better available satellite data.

2 Related Work

The detection and classification of urban trees using remote sensing imagery is a rapidly evolving field, characterized by a transition from traditional algorithms to advanced machine learning techniques. A comprehensive overview is provided by Zheng et al. [11], who review various methods for individual tree crown detection and delineation from optical imagery, highlighting the growing reliance on deep learning architectures. Building on this methodological shift, Durgut et al. [12] demonstrate the benefits of integrating traditional image processing methods with modern deep learning approaches to further enhance the accuracy of tree crown detection in satellite images.

A significant portion of recent research focuses on highly detailed classification tasks utilizing very high-resolution (VHR) imagery. For instance, Sicard et al. [13] explore object-based classification techniques to accurately distinguish different urban plant species. Similarly, Wenger et al. [14] focus on improving urban tree species classification by combining VHR satellite data with various machine learning models. While these approaches achieve high precision at the individual tree level, they often depend on commercial or less frequently updated satellite data, which can limit the scalability and temporal analysis required for continuous urban climate monitoring.

Addressing the need for scalable and frequently updated data, recent studies have increasingly adopted Sentinel-2 imagery for forestry and urban green space applications. Notably, Freudenberg et al. [15] introduce a dedicated Sentinel-2 machine learning dataset specifically designed for tree species classification in Germany. Their work underscores the viability of using freely available, multi-spectral satellite data for large-scale vegetation mapping.

Furthermore, recent advancements highlight the critical role of multi-temporal analysis in vegetation mapping. Because the spectral signature of trees changes significantly throughout the year due to phenological cycles, relying on single-date imagery often leads to misclassifications. Grabska et al. [16] demonstrate that incorporating the entire Sentinel-2 time series—capturing leaf-on, leaf-off, and flowering periods—greatly enhances the ability of models to differentiate vegetation types. Integrating these temporal signals into deep learning architectures, such as through convolutional temporal attention networks, allows models to prioritize the most meaningful seasonal features for precise identification [10, 16].

Overall, while existing literature predominantly focuses either on VHR imagery for exact crown delineation or on traditional models for general species classification, our project bridges these areas. By leveraging the temporal resolution of Sentinel-2 data [15, 16] combined with a temporal attention U-Net architecture [10], this study aims to provide a scalable and robust solution for urban tree detection that does not rely on costly VHR imagery.

3 Data Sources

4 Methodology

4.1 BEAM

Figure 1 presents the BEAM representation of the proposed tree detection system, illustrating the data sources, training and inference processes, model architecture, evaluation pipeline, and associated risks.

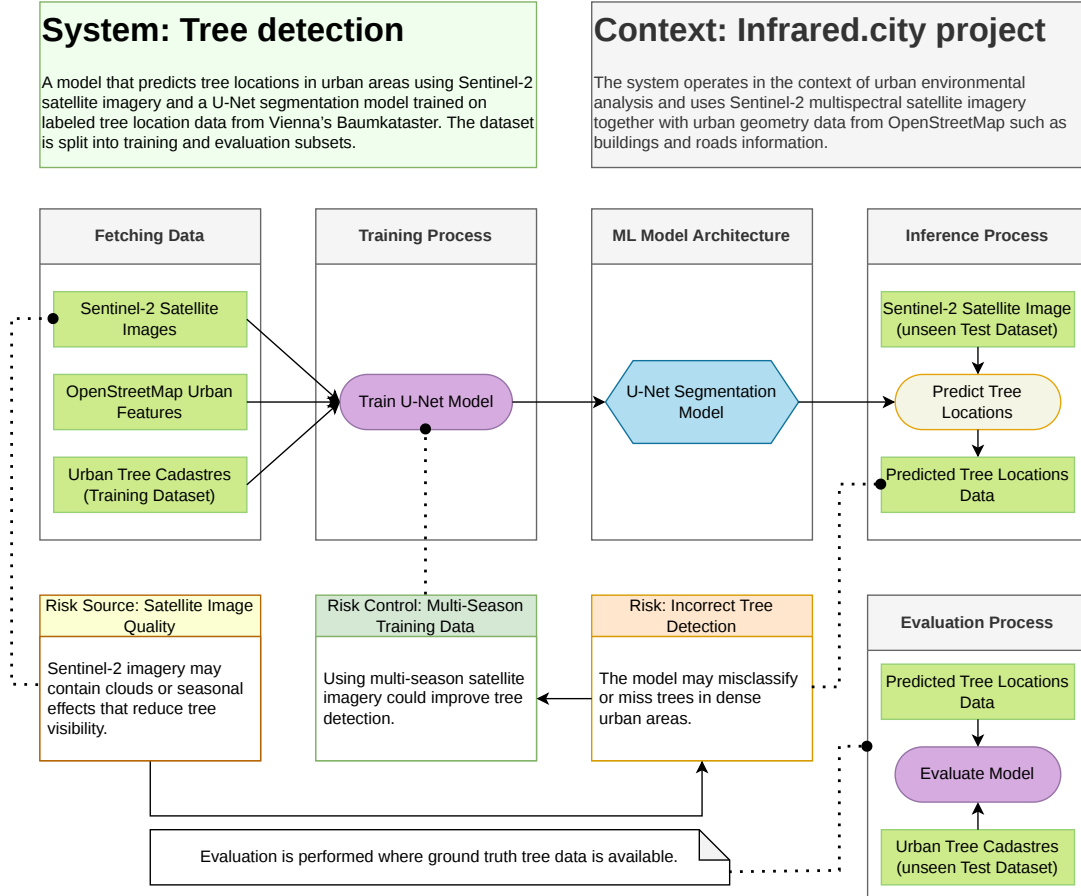


Figure 1: BEAM model of the proposed tree detection system.

5 Implementation

6 Preliminary Results

7 Preliminary Discussion

8 Preliminary Conclusion

References

- [1] European Commission, “Increasing tree coverage to 30% in european cities could reduce deaths linked to urban heat island effect,” 2023. Abgerufen: 2026-03-15.
- [2] Woodland Trust, “Benefits of urban trees,” 2023. Abgerufen: 2026-03-15.
- [3] Marco, Rafael, Osman, and Hanno, “Flood risk analysis for infrared city: A data-driven analysis of satellite imagery,” final report, Vienna University of Economics and Business (WU), Data Science Lab, February 2026.
- [4] T. Gotschim, L. R. Hadrys, and A. Cherkaskyy, “Green space data enhancement through computer vision,” final report, Vienna University of Economics and Business (WU), Data Science Lab, January 2026.
- [5] H. Gong, Q. Sun, C. Fang, L. Sun, and R. Su, “Treedetector: Using deep learning for the localization and reconstruction of urban trees from high-resolution remote sensing images,” *Remote Sensing*, vol. 16, no. 3, p. 524, 2024.
- [6] Copernicus Programme, “Copernicus data space browser,” 2025. Abgerufen: 2026-03-15.
- [7] data.gv.at, “Baumkataster,” 2025. Abgerufen: 2026-03-15.
- [8] A. I. Aramendia, “Decoding the u-net: A complete guide,” 2024. Abgerufen: 2026-03-15.
- [9] O. Pesek, L. Krisztian, M. Landa, M. Metz, and M. Neteler, “Convolutional neural networks for road surface classification on aerial imagery,” *PeerJ Computer Science*, vol. 10, p. e2571, 2024.
- [10] V. S. F. Garnot and L. Landrieu, “Panoptic segmentation of satellite image time series with convolutional temporal attention networks,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 4872–4881, 2021.
- [11] J. Zheng, S. Yuan, W. Li, H. Fu, and L. Yu, “A review of individual tree crown detection and delineation from optical remote sensing images,” 2023.
- [12] O. Durgut, B. Kallfelz-Sirmacek, and C. Ünsalan, “Integrating traditional and deep learning methods to detect tree crowns in satellite images,” 2025.
- [13] P. Sicard, F. Coulibaly, M. Lameiro, V. Araminiene, A. De Marco, B. Sorrentino, A. Anav, J. Manzini, Y. Hoshika, B. B. Moura, *et al.*, “Object-based classification of urban plant species from very high-resolution satellite imagery,” *Urban Forestry & Urban Greening*, vol. 81, p. 127866, 2023.
- [14] R. Wenger, C. Bressant, L. Roettele, G. Forestier, and A. Puissant, “Improving urban tree species classification with high resolution satellite imagery and machine learning,” in *IEEE Conference Publication*, IEEE, 2024.
- [15] M. Freudenberg, S. Schnell, and P. Magdon, “A sentinel-2 machine learning dataset for tree species classification in germany,” *Earth System Science Data*, vol. 17, pp. 351–367, 2025.

- [16] E. Grabska, P. Hostert, J. Pflugmacher, and K. Ostapowicz, “Forest stand species mapping using the Sentinel-2 time series,” *Remote Sensing*, vol. 11, no. 10, p. 1197, 2019.

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